

Name: _____

SIMULTANEOUS EQUATIONS, (SOLVING BY SUBSTITUTION)

Date: _____

LO: I can solve simultaneous linear equations by substituting one equation into the other.

The substitution method

The **substitution method** involves rearranging one equation to make a variable the subject, then **replacing** that variable in the other equation.

Strategy: Choose the equation where one variable is easiest to isolate (coefficient of 1). Substitute into the other equation. Solve, then find the second variable.

Quick examples

●	$y = 2x + 1$ into $3x + y = 11$: $3x + (2x+1) = 11$	direct substitution
●	$x = 5 - y$ into $2x + 3y = 13$: $2(5-y) + 3y = 13$	rearrange first
●	Best when one variable already has coefficient 1	easy to rearrange
●	$y = 2x + 1$ into $3x + y = 11$ gives $3x + 2x + 1 = 11$ but answer is $x = 3$	$5x + 1 = 11$, $x = 2$

Key idea

Make one variable the subject of one equation, then substitute that expression into the other equation.

$$y = \textcircled{3x - 1}; 2x + y = 9 \quad 2x + \textcircled{3x - 1} = 9$$

Replace y with the expression from the first equation

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - y already the subject

Solve $y = 2x + 1$ and $3x + y = 11$

Substitute $y = 2x + 1$ into equation

$$2: 3x + (2x + 1) = 11$$

$$5x + 1 = 11 \quad 5x = 10 \quad x = 2$$

$$y = 2(2) + 1 = 5$$

$$\boxed{x = 2, y = 5}$$

When y is already isolated, substitute directly.

Example 2 - rearrange first

Solve $x + y = 8$ and $2x + 3y = 21$

From eq 1: $y = 8 - x$

$$\text{Substitute: } 2x + 3(8 - x) = 21 \quad 2x + 24 - 3x = 21$$

$$-x = -3, \text{ so } x = 3; y = 8 - 3 = 5$$

$$\boxed{x = 3, y = 5}$$

Rearrange the simpler equation to make y the subject.

Example 3 - x as the subject

Solve $x = 3y - 2$ and $2x + y = 10$

Substitute: $2(3y - 2) + y = 10$

$$6y - 4 + y = 10 \quad 7y = 14 \quad y = 2$$

$$x = 3(2) - 2 = 4$$

$$\boxed{x = 4, y = 2}$$

If x is easily isolated, substitute the x -expression.

Example 4 - checking the solution

Solve $y = x - 3$ and $2x + 5y = 23$. Verify.

Substitute: $2x + 5(x - 3) = 23 \quad 7x - 15 = 23 \quad 7x = 38 \dots$ let me recalculate

$$2x + 5x - 15 = 23 \quad 7x = 38 \dots x = \frac{38}{7} \dots$$

Hmm, let me use integer-friendly:

$$y = x - 3 \text{ and } 2x + 5y = 19: 2x + 5(x - 3) = 19,$$

$$7x - 15 = 19, 7x = 34 \dots \text{ use } y = x - 3 \text{ and } 4x + y = 17:$$

$$4x + x - 3 = 17, 5x = 20, x = 4,$$

$$y = 1$$

$$\text{Check: } 2(4) + 5(1) = 8 + 5 = 13 \dots \text{ Let me use: } y = x - 3 \text{ and } 3x + 2y = 19:$$

$$3(4) + 2(1) = 12 + 2 = 14 \dots$$

YOUR TURN – PRACTICE (deliberate practice)

1. y is already the subject

Solve by substitution.

- a) $y = x + 3$ and $2x + y = 12$
- b) $y = 3x$ and $x + y = 8$
- c) $y = 2x - 1$ and $x + y = 8$
- d) $y = x + 5$ and $3x + y = 13$

2. Rearrange one equation

Rearrange, then substitute.

- a) $x + y = 10$ and $3x + 2y = 26$
- b) $x + y = 7$ and $2x + 5y = 20$
- c) $x + 2y = 9$ and $3x + y = 12$
- d) $x - y = 1$ and $3x + 2y = 18$

3. x as the subject

Make x the subject, then substitute.

- a) $x = 2y$ and $3x + y = 14$
- b) $x = y + 4$ and $2x + 3y = 18$
- c) $x = 3y - 1$ and $x + 2y = 9$
- d) $x = 10 - 2y$ and $3x + y = 15$

4. More complex

Solve by substitution.

- a) $2y = x + 3$ and $3x + 4y = 21$
- b) $x = 2y - 5$ and $4x + 3y = 9$
- c) $y = 3x - 7$ and $2x + y = 3$
- d) $x + 3y = 11$ and $2x - y = 1$

5. Mixed substitution

Solve using substitution.

- a) $y = 4x - 3$ and $2x + y = 9$
- b) $x + y = 5$ and $3x - 2y = 5$
- c) $y = x - 4$ and $5x + 2y = 27$
- d) $2x + y = 11$ and $x - y = 1$
- e) $x = 2y - 3$ and $3x + 4y = 21$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Substitution works by replacing a variable with an equivalent expression ☐

Correct answer: _____

Reason: _____

b) You must always make y the subject ☐

Correct answer: _____

Reason: _____

c) After solving for one variable, substitute back to find the other ☐

Correct answer: _____

Reason: _____

d) Substitution only works when y is already the subject ☐

Correct answer: _____

Reason: _____

e) You should always check your answer in both original equations ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Solve: $3x + 2y = 20$ and $x = y + 2$. A shop sells pens at £x and notebooks at £y. If 3 pens and 2 notebooks cost £20, and a pen costs £2 more than a notebook, find the price of each.

Expression: _____

Simplified: _____

b) The perimeter of a rectangle is 34 cm. The length is 3 cm more than the width. Write two equations and solve by substitution to find the length and width.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

